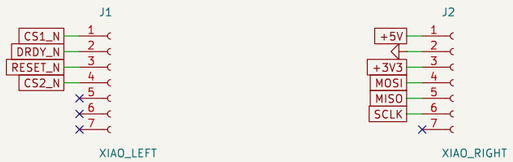
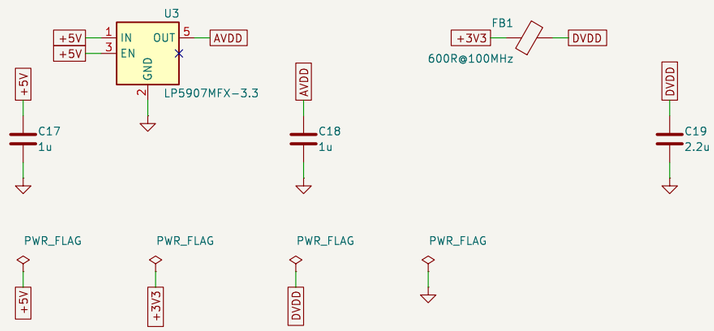
NOT A MEDICAL DEVICE. No isolation: run from a battery / power bank only while worn.

XIAO ESP32-S3 sockets (USB-C at the top)

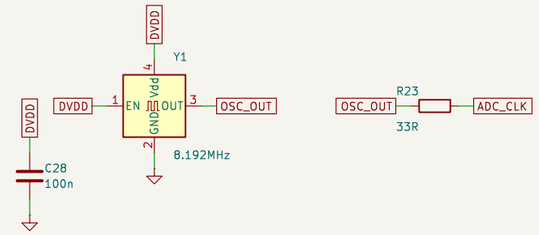


D0 CS1 D1 DRDY D2 RESET D3 CS2
D4–D7 unused D8 SCLK D9 MISO D10 MOSI

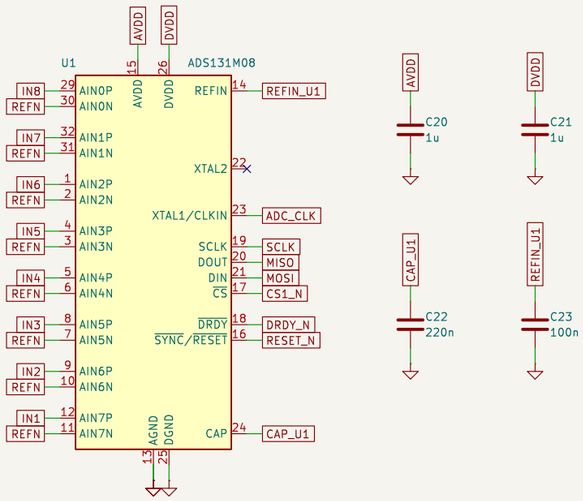
Power: quiet 3.3 V analog rail, filtered digital rail



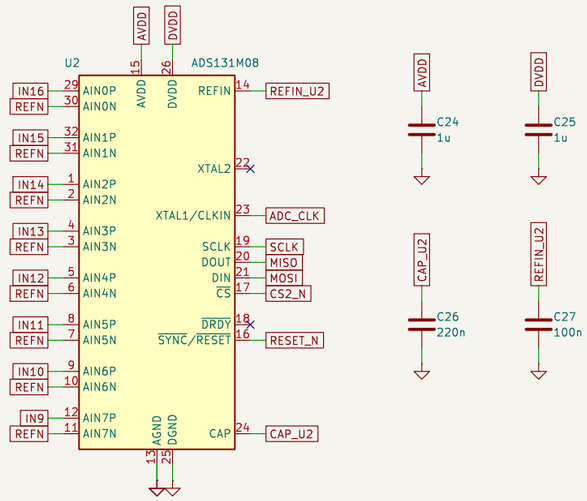
Clock: 8.192 MHz to both converters



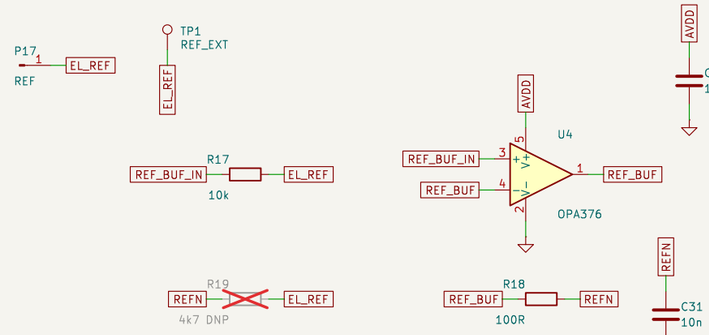
ADC A: channels 1–8



ADC B: channels 9–16

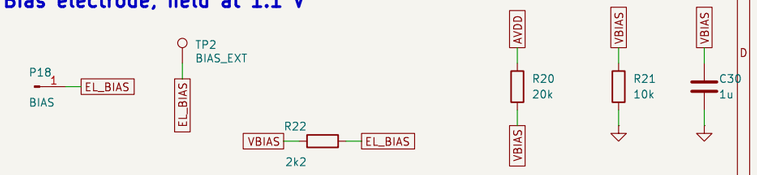


Reference electrode and unity-gain buffer



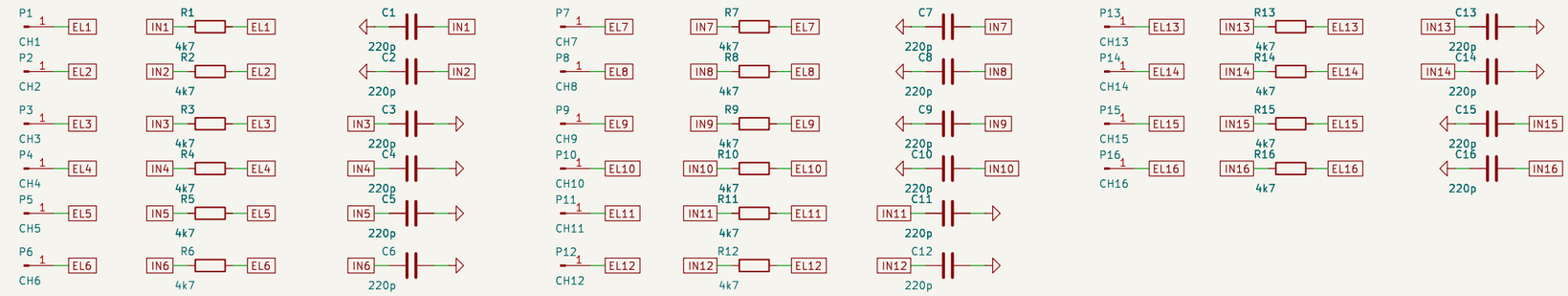
R18 fitted, R19 not fitted: buffered reference (default).
R19 fitted, R18 removed, C31 changed to 220 pF: electrode drives the reference node directly.

Bias electrode, held at 1.1 V



Remove R20 to hold the bias electrode at ground (then use the direct reference: the buffer needs the 1.1 V bias to work).

Electrodes and input filters: pogo pin -> 4k7 -> ADC input, 220 pF to ground



Untested revision. Not a medical device.
Generated from hardware/scripts/design.py. Edit that file, not this one.
Open hardware, CERN-OHL-P-2.0

Sheet: /
File: limpet16.kicad_sch

Title: Limpet-16: 16-channel dry-electrode EEG front end

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 9.0.9.1		Id: 1/1